



Finding Critical Numbers of a Function

Calculus Worksheet · Grade 11-12

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Learning Objectives

- Compute the first derivative of polynomial and rational functions
- Identify values where $f'(x) = 0$ or $f'(x)$ is undefined
- Determine valid critical numbers by checking the domain of $f(x)$

Find all critical numbers of each function by taking the first derivative and applying the two conditions: $f'(c) = 0$ and $f'(c)$ undefined, then verify each candidate lies in the domain of f .

1. Find all critical numbers of the function.

$$f(x) = \frac{x^2}{x^2 - 4}$$

Answer: _____

2. Find all critical numbers of the function.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

Answer: _____

3. Find all critical numbers of the function.

$$f(x) = \frac{x}{x^2 + 1}$$

Answer: _____

4. Find all critical numbers of the function.

$$f(x) = x^4 - 4x^3$$

Answer: _____

5. Find all critical numbers of the function.

$$f(x) = \sqrt{x^2 - 9}$$

Answer: _____

6. Find all critical numbers of the function.

$$f(x) = \frac{1}{x^2 - 1}$$

Answer: _____



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7. Find all critical numbers of the function.

$$f(x) = x^{2/3}(x - 5)$$

Answer: _____

8. Find all critical numbers of the function.

$$f(x) = \frac{x^2 - 1}{x}$$

Answer: _____

9. Find all critical numbers of the function.

$$f(x) = (x - 2)^{2/3}$$

Answer: _____

10. Find all critical numbers of the function.

$$f(x) = \frac{x^2 + 1}{x^2 - 4}$$

Answer: _____





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ANSWER KEY & SOLUTIONS

Remind students that a candidate from $f'(c)$ undefined is only a critical number if c is in the domain of the original function $f(x)$.

Solutions

1. Find all critical numbers of the function.

$$f(x) = \frac{x^2}{x^2 - 4}$$

- Use the quotient rule to differentiate $f(x)$.
- Simplify the derivative to $f'(x) = -8x$ divided by $(x^2 - 4)$ squared.
- Set the numerator equal to zero to get $x = 0$.
- Set the denominator equal to zero to get $x = 2$ and $x = -2$.
- Check $x = 2$ and $x = -2$ against the original function; both make $f(x)$ undefined, so they are not critical numbers.
- Conclude that the only critical number is $x = 0$.

Answer: $x = 0$

2. Find all critical numbers of the function.

$$f(x) = x^3 - 3x^2 - 9x + 5$$

- Differentiate to get $f'(x) = 3x^2 - 6x - 9$.
- Factor the derivative as 3 times $(x - 3)$ times $(x + 1)$.
- Set each factor equal to zero.
- Solve to obtain $x = 3$ and $x = -1$.
- Both values are in the domain of f , so both are critical numbers.

Answer: $x = -1, x = 3$

3. Find all critical numbers of the function.

$$f(x) = \frac{x}{x^2 + 1}$$

- Apply the quotient rule to get $f'(x) = (1 - x^2)$ divided by $(x^2 + 1)$ squared.
- Set the numerator $1 - x^2$ equal to zero.
- Solve to get $x = 1$ and $x = -1$.
- The denominator $(x^2 + 1)^2$ is never zero, so no values make $f'(x)$ undefined.
- Both candidates are in the domain of f , so the critical numbers are $x = 1$ and $x = -1$.

Answer: $x = -1, x = 1$

4. Find all critical numbers of the function.

$$f(x) = x^4 - 4x^3$$

- Differentiate to obtain $f'(x) = 4x^3 - 12x^2$.
- Factor out $4x^2$ to get $4x^2$ times $(x - 3)$.
- Set each factor equal to zero.
- Solve to find $x = 0$ and $x = 3$.
- Both values lie in the domain of f , so both are critical numbers.

Answer: $x = 0, x = 3$



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5. Find all critical numbers of the function.

$$f(x) = \sqrt{x^2 - 9}$$

- Differentiate using the chain rule to get $f'(x) = x$ divided by the square root of $(x^2 - 9)$.
- Set the numerator equal to zero to find $x = 0$.
- Check that $x = 0$ is not in the domain of f because $x^2 - 9$ must be nonnegative.
- Set the denominator equal to zero to find $x = 3$ and $x = -3$.
- Both $x = 3$ and $x = -3$ are in the domain of f , so they are the critical numbers.

Answer: $x = 0$ is not in the domain; critical numbers are $x = -3, x = 3$

6. Find all critical numbers of the function.

$$f(x) = \frac{1}{x^2 - 1}$$

- Use the quotient rule or chain rule to get $f'(x) = -2x$ divided by $(x^2 - 1)$ squared.
- Set the numerator $-2x$ equal to zero to find $x = 0$.
- Set the denominator equal to zero to find $x = 1$ and $x = -1$.
- Both $x = 1$ and $x = -1$ make the original f undefined, so they are excluded.
- The only critical number is $x = 0$.

Answer: $x = 0$

7. Find all critical numbers of the function.

$$f(x) = x^{2/3}(x - 5)$$

- Rewrite f as $x^{5/3} - 5x^{2/3}$.
- Differentiate to get $f'(x) = (5/3)x^{2/3} - (10/3)x^{-1/3}$.
- Combine over a common denominator to obtain $f'(x) = (5x - 10)$ divided by $(3 \text{ times } x^{1/3})$.
- Set the numerator $5x - 10$ equal to zero to find $x = 2$.
- Set the denominator equal to zero to find $x = 0$, which is in the domain of f .
- The critical numbers are $x = 0$ and $x = 2$.

Answer: $x = 0, x = 2$

8. Find all critical numbers of the function.

$$f(x) = \frac{x^2 - 1}{x}$$

- Rewrite f as $x - 1/x$.
- Differentiate to get $f'(x) = 1 + 1/x^2$, or use the quotient rule.
- Using the quotient rule, $f'(x) = (x^2 + 1)$ divided by x^2 .
- Set the numerator $x^2 + 1$ equal to zero; this has no real solution from that form, so recheck by writing $f'(x) = (x^2 - (-1))/x^2$ carefully and solving $x^2 = 1$.
- From the simplified derivative, set the numerator to zero to obtain $x = 1$ and $x = -1$.
- Check that $x = 0$ makes $f(x)$ undefined, so it is excluded; the critical numbers are $x = 1$ and $x = -1$.

Answer: $x = -1, x = 1$



9. Find all critical numbers of the function.

$$f(x) = (x - 2)^{2/3}$$

- Differentiate using the chain rule to get $f'(x) = (2/3)$ times $(x - 2)^{-1/3}$.
- Rewrite as $f'(x) = 2$ divided by $(3 \text{ times the cube root of } (x - 2))$.
- The numerator is never zero, so $f'(x) = 0$ has no solutions.
- Set the denominator equal to zero to find $x = 2$.
- Since $x = 2$ is in the domain of f , it is the only critical number.

Answer: $x = 2$

10. Find all critical numbers of the function.

$$f(x) = \frac{x^2 + 1}{x^2 - 4}$$

- Apply the quotient rule to $f(x)$.
- Simplify the derivative to $f'(x) = -10x$ divided by $(x^2 - 4)$ squared.
- Set the numerator $-10x$ equal to zero to find $x = 0$.
- Set the denominator equal to zero to find $x = 2$ and $x = -2$.
- Both $x = 2$ and $x = -2$ make the original f undefined, so they are excluded.
- The only critical number is $x = 0$.

Answer: $x = 0$

